

PRAGNA MAMIDIPAKA

Curriculum Vitae

Ph.D. Candidate, Carnegie Mellon University
Department of Electrical and Computer Engineering
Pittsburgh, PA, USA

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Personal Website

BRIEF BIOGRAPHY

My research focuses on systems observability, with the goal of making complex systems easier to understand, diagnose, and improve. I am interested in the operating-system, networking, and runtime support needed to observe and explain system behavior. My broader goal is to connect observations across system layers and enable adaptive diagnosis by both humans and AI agents. I have applied this perspective to data center congestion control, observing and modeling algorithm execution to guide the selection of workloads for evaluation. My research has been awarded with the Prabhu and Poonam Goel Graduate Fellowship and the Carnegie Institute of Technology Dean's Fellowship.

EDUCATION

Ph.D., Electrical and Computer Engineering Carnegie Mellon University, Pittsburgh, PA Advisor: Prof. Theophilus Benson	2024–2029 (expected) <i>Grade: 4.00/4</i>
B.Tech., Electrical Engineering Indian Institute of Technology Hyderabad, India Minor in Artificial Intelligence <i>President of India Gold Medallist</i>	2020–2024 <i>Grade: 9.75/10</i>
High School and Senior Secondary Education City International School, Pune, India; Science discipline <i>Ranked first in the school in the CBSE board examination.</i>	2007–2020

HONORS AND AWARDS

Prabhu and Poonam Goel Graduate Fellowship Awarded by the Carnegie Mellon University ECE Department.	2026
Carnegie Institute of Technology Dean's Fellow Full tuition support and stipend for one year of Ph.D. study.	2024–2025
President of India Gold Medal Highest overall CGPA among all graduating B.Tech. students at IIT Hyderabad.	2024
Institute Silver Medal Highest CGPA in the Electrical Engineering Department at IIT Hyderabad.	2024
Academic Excellence Award Recognized for outstanding academic performance.	2024; 2022
Excellence in Innovation Award Among the top 5 of 1,300 participants in the Indo-Japanese Road2Shine program.	2022
All India Rank 3418, JEE-Advanced	2020
CBSE Merit Certificate Top 0.1% in Grade 10 Science among 1.7 million candidates.	2018

PUBLICATIONS AND PREPRINTS

Systems and Networking

- Animesh Singh, K Shiv Kumar, S. VenkataKeerthy, Pragna Mamidipaka, R V B R N Aaseesh, Sayandeep Sen, Palanivel Kodeswaran, Theophilus A. Benson, Ramakrishna Upadrasta, Praveen Tammana. *Yaksha-Prashna: Understanding eBPF Bytecode Network Function Behavior*. ACM CoNEXT, 2026. (Accepted.)
- Pragna Mamidipaka, Srikanth Sundaresan, Theophilus A. Benson. *Towards Truly Burst-Aware Evaluation of Data Center Congestion Control*. Asia-Pacific Workshop on Networking (APNet), 2026.
- Pragna Mamidipaka, Theophilus A. Benson. *Apiary: Distributed Programming and Lifecycle Management for eBPF*. N2Women Workshop at SIGCOMM, 2025.

Pulsar Astronomy

- **The Indian Pulsar Timing Array Data Release 2: I. Dataset and Timing Analysis.** *Publications of the Astronomical Society of Australia*, 2025.
- **Low-frequency pulse-jitter measurement with the uGMRT I: PSR J0437–4715.** *Publications of the Astronomical Society of Australia*, 2024.
- **Application of Efron-Petrosian method to radio pulsar fluxes.** *Journal of Cosmology and Astroparticle Physics*, 2024.
- **Do Pulsar and Fast Radio Burst dispersion measures obey Benford’s law?** *Astroparticle Physics*, 2023.

PROFESSIONAL EXPERIENCE

Software Developer, Arcesium

May 2023–Jul 2023

Hyderabad, India

Developed a CI/CD workflow analysis application with dashboards in Apache Superset and Grafana.

Undergraduate Researcher, University of Tokyo

Dec 2022

Tokyo, Japan; advisor: Prof. Kunihiko Sadakane

Studied succinct encoding of ordinal trees using tree covering during a research internship.

Student Member, Indian Pulsar Timing Array

Jun 2022–Jun 2024

Hyderabad, India

Contributed to the InPTA Consortium’s Data Reduction team. Worked with the National Centre for Radio Astrophysics to collate legacy data from the upgraded Giant Metrewave Radio Telescope (uGMRT).

5G Testbed Intern, Wisig Networks

Jan 2022–Jun 2022

Hyderabad, India

Developed a module to transmit Downlink Control Information (DCI) from a gNB base station to user equipment, gaining experience with L2/L3 protocol stack development.

PROFESSIONAL SERVICE

Artifact Evaluation Committee Member, CAIS 2026

Served on the Artifact Evaluation Committee for the first ACM Conference on AI and Agentic Systems.

Shadow Program Committee Member, IMC 2026

Shadow Program Committee Member, ACM CoNEXT 2025

Evaluator, Meeting of the Minds, Carnegie Mellon University 2025

Reviewed senior undergraduate research projects at CMU's annual research symposium.

TEACHING AND COMMUNITY OUTREACH

K–12 Community Outreach 2026

Focus on making engineering more accessible and equitable in K–12 spaces and re-envisioning engineering education pedagogy.

Yoga Instructor, Carnegie Mellon University 2024–Present

Teach weekly group yoga sessions through CMU's GroupX fitness program.

Volunteer Teacher, National Service Scheme, India 2022–2024

Taught Grade 10 mathematics at rural schools and led motivational lectures as part of the Aksharamaala program.